

Differentially Private Preference Data Synthesis for Large Language Model Alignment

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Abstract

Preference alignment is a central post-training mechanism for large language models (LLMs), yet its reliance on real human preference data exposes sensitive prompts, responses, and human judgments. We introduce **DPPrefSyn**, a differentially private (DP) algorithm for synthesizing preference data that supports privacy-preserving LLM alignment. DPPrefSyn provides a principled synthesis pipeline grounded in the Bradley–Terry preference model and in the geometric structure of pairwise preference representations. The algorithm first learns DP-protected preference models from private comparisons, and then combines these models with public prompts to construct synthetic preference pairs. Its design captures heterogeneous human preferences through the shared linear structure of per-cluster reward models and improves private learning accuracy through DP Principal Component Analysis (DP-PCA). Experiments across question answering and summarization benchmarks show that DPPrefSyn delivers competitive alignment utility under strong DP guarantees, establishing DP synthetic preference data as a practical substrate for privacy-preserving alignment across multiple post-training methods. To the best of our knowledge, this is the first work to generate DP synthetic preference data for LLM alignment. Our code will be released upon acceptance.

1. Introduction

Preference alignment methods, including RLHF (Stiennon et al., 2020) and DPO (Rafailov et al., 2024), have become a primary mechanism for adapting LLMs to human expectations. These methods are organized around pairwise preference data: annotators compare two candidate responses

to a shared prompt and identify the response that better satisfies the target criterion. The resulting comparisons shape post-training by increasing the model’s tendency to rank preferred responses above less preferred alternatives (Ziegler et al., 2019; Ouyang et al., 2022; Bai et al., 2022). Because this paradigm reliably improves interactive model behavior, it now supports deployed and research systems for chat assistance (Achiam et al., 2023), mathematical reasoning (Shao et al., 2024), and code generation (Shen et al., 2023).

The same data that gives preference alignment its empirical strength also creates a substantial privacy problem. Preference datasets often contain real user prompts and human judgments; prompts may disclose health, identity, or other sensitive information, while feedback can reveal private beliefs, preferences, and behavioral regularities (Li et al., 2023; Yu et al., 2024). Differential privacy (DP) (Dwork et al., 2006b) provides a rigorous framework for controlling such disclosure, and recent DP alignment studies have made important progress (Chowdhury et al., 2024; Yu et al., 2024; Wu et al., 2023a). Nevertheless, existing approaches leave a key gap. Some privatize only a single component of the preference record, such as user prompts (Yu et al., 2024) or preference labels (Chowdhury et al., 2024; Zhang et al., 2025), while leaving the remaining components exposed to inference. Others are tied to a particular post-training procedure, including RLHF (Wu et al., 2023a), and therefore do not directly provide a reusable privacy-preserving data interface for newer objectives such as DPO. Moreover, approaches that train directly on limited private preference data inherit the cost, scarcity, and scalability constraints of human annotation, which can sharply constrain alignment utility under strong privacy budgets. These limitations motivate the central question of this work:

Is it possible to achieve superior preference alignment while ensuring thorough privacy protection and maintaining compatibility with emerging alignment methodologies?

This paper answers that question affirmatively by constructing DP synthetic preference data that protects the underlying private comparison dataset while remaining directly usable by downstream alignment algorithms. The work makes three connected contributions:

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- We introduce DPPrefSyn (Figure 1), a DP preference-data synthesis algorithm for LLM alignment. DPPrefSyn is grounded in the Bradley–Terry preference model and in the geometry of pairwise preference embeddings. Its design integrates three mechanisms: 1) *Capturing heterogeneous human preferences through shared linear reward structure and clustering*. The Bradley–Terry model and a linear reward parameterization enable DPPrefSyn to cluster comparisons by embedding differences between prompt–preferred and prompt–dispreferred pairs, and to represent each cluster with a shared reward function. 2) *Improving private learning efficiency through DP-PCA*. DP-PCA maps high-dimensional embeddings into a lower-dimensional subspace that retains the preference signal required for private clustering and reward learning. 3) *Preserving privacy budget through public prompts*. Public prompts elicit candidate responses, while DP reward models select synthetic preference pairs, concentrating privacy expenditure on private preference modeling rather than prompt generation.
- We give a privacy analysis proving that DPPrefSyn satisfies (ϵ, δ) -DP. The analysis allocates privacy budget ϵ_0 to DP-PCA (Amin et al., 2019), ϵ_1 to DP-KMeans (Su et al., 2016), and the remaining budget to per-cluster reward-model training with DP-SGD. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) provides tight composition for the DP-SGD component. By DP post-processing, the synthesized preference dataset can then be reused with multiple alignment methods and LLMs without additional privacy expenditure.
- We evaluate DPPrefSyn on OpenAssistant (Köpf et al., 2023) and Anthropic-HH (Bai et al., 2022) question answering tasks and on TL;DR summarization (Stiennon et al., 2020). Across these settings, DPPrefSyn produces DP synthetic preference data with strong utility under privacy constraints. For example, at $\epsilon = 2$ on Anthropic-HH, DPO fine-tuning of Pythia-2.8B with DPPrefSyn data obtains a GPT-4o win rate of 56.48%, compared with 37.02% for fine-tuning on real data. The results establish that DP synthetic preference data can support effective alignment under strong privacy guarantees and can be used with both DPO and RLHF-style downstream alignment procedures.

2. Related Work

Differentially Private Alignment of LLMs. Wu et al. (2023a) introduce a DP framework for reinforcement-learning-based LLM alignment by adapting PPO to the DP setting. This work provides an important privacy mechanism for PPO, but it remains *limited to PPO* even as preference alignment increasingly relies on alternative objectives. Chowdhury et al. (2024) study reward estimation from

preference-based feedback under label-DP (Ghazi et al., 2021), and Zhang et al. (2025) propose AUP-RLHF as a user-level label-DP framework for RLHF. These methods protect preference labels, but they *do not address the sensitivity of prompts or responses themselves*. They also remain *constrained by the limited private user preference data* available for alignment, because human preference annotations are expensive to collect at scale. Yu et al. (2024) instead protect sensitive user *instructions* by generating DP synthetic *instruction* data for supervised fine-tuning, while preference labels remain outside the privacy target. DPPrefSyn fills this gap by protecting the full preference record—prompts, responses, and labels—through synthesis of a DP preference dataset from private user data. Because the released object is a DP dataset rather than a method-specific model update, it can be reused across LLMs and preference alignment methods for privacy-preserving annotation expansion and model fine-tuning.

Beyond preference alignment, another line of work studies private LLM fine-tuning with general labeled data rather than pairwise comparisons (Yu et al., 2021; Li et al., 2021; Chen et al., 2024; Tang et al., 2024; Zhang et al., 2023; He et al., 2022). Recent studies also protect in-context prompts used for in-context learning (Duan et al., 2023; Tang et al., 2023; Gao et al., 2024; Hong et al., 2023; Wu et al., 2023b). These settings are complementary to the problem addressed here, where the private object is a structured preference comparison and the synthesized output must remain compatible with preference-based post-training.

Differentially Private Synthetic Text Generation. DPPrefSyn also builds on the broader literature on DP synthetic text generation, where two representative paradigms have emerged. The first privately fine-tunes a language model on sensitive data and then samples synthetic text from the DP-trained model (Yue et al., 2023; Mattern et al., 2022; Miresghallah et al., 2023; Carranza et al., 2023; Yu et al., 2024; Wang et al., 2024; Ochs & Habernal, 2024; Tan et al., 2025; Carranza et al., 2024). For example, Yu et al. (2024) privately fine-tune LLaMA on private instruction data and use the resulting model to generate synthetic text for supervised instruction tuning.

Because DP fine-tuning can be computationally expensive or inaccessible for non-public language models, a second paradigm generates DP synthetic data through LLM APIs (Lin et al., 2023; Xie et al., 2024; Wu et al., 2024; Hou et al., 2024). These methods typically prompt an LLM to generate candidates and then select outputs that resemble private data through a privacy-preserving selection procedure. For instance, Xie et al. (2024) propose Aug-PE, which iteratively refines LLM-generated samples using embedding-space similarity to private data. *DPPrefSyn extends this API-based synthetic-data perspective from single-text syn-*

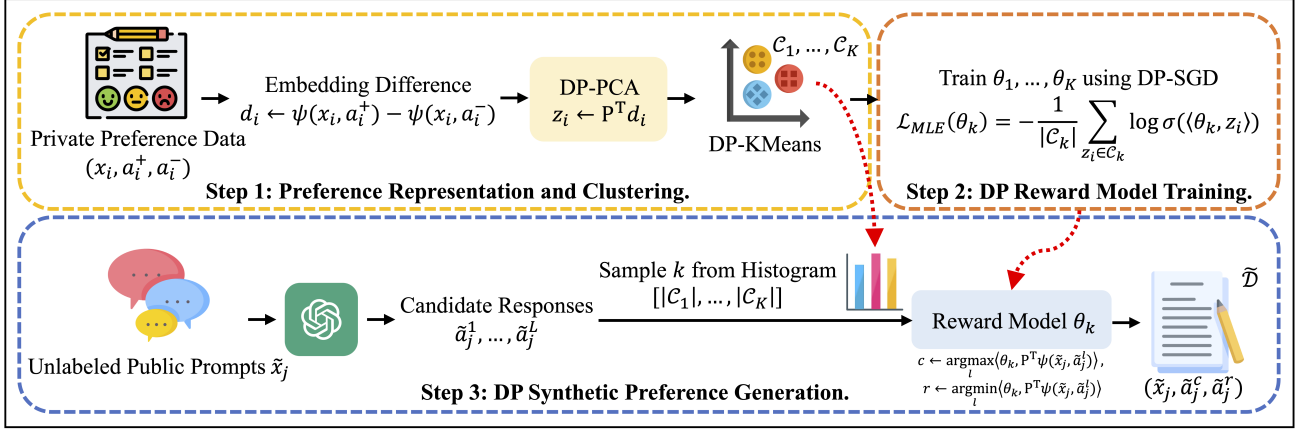


Figure 1. Overview of DPPrefSyn. The algorithm synthesizes DP preference data in three stages: 1) represent private comparisons as embedding differences and cluster them with DP-PCA and DP-KMeans, 2) train a DP reward model for each cluster with DP-SGD, and 3) generate synthetic preference pairs from public prompts using the DP-protected distribution over learned reward functions.

thesis to preference alignment, where the protected and reproduced object is a structured pairwise comparison.

Diversity in Human Preferences. Human preference data is not monolithic: preferences vary across annotators and user groups (Denton et al., 2021; Aroyo et al., 2023a,b; Chakraborty et al., 2024). Relevant sources of variation include sociodemographic differences, personal beliefs and biases, domain expertise, and ambiguity in natural language interpretation (Sandri et al., 2023; Vogels, 2021). This observation has motivated preference-alignment methods that adapt to different user values or groups (Lee et al., 2024b; Poddar et al., 2024; Singh et al., 2025). For example, Singh et al. (2025) propose a meta-learning framework in which an LLM adapts to individual users from a small number of preference examples. DPPrefSyn incorporates this heterogeneity into the synthesis mechanism itself: it generates DP synthetic preference data that captures the underlying diversity of human feedback, and the ablations in Section 5.3 show that preserving this structure improves alignment relative to variants that ignore it.

3. Problem Definition

Preference Alignment of LLMs. We study LLM preference alignment from pairwise human feedback, represented by a preference dataset $\mathcal{D} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$. Each record consists of a prompt x_i and two candidate responses a_i^+, a_i^- , with a_i^+ denoting the response preferred by human annotators over a_i^- .

Following prior work on reinforcement learning from human feedback (RLHF) and direct preference optimization (DPO) (Ouyang et al., 2022; Zhu et al., 2023; Rafailov et al., 2024; Liu et al., 2023), we formalize human comparisons

with the Bradley–Terry model (Bradley & Terry, 1952):

$$\begin{aligned} \mathbb{P}[a_i^+ \succ a_i^- \mid x_i] &= \frac{\exp(r^*(x_i, a_i^+))}{\exp(r^*(x_i, a_i^+)) + \exp(r^*(x_i, a_i^-))} \\ &= \sigma(r^*(x_i, a_i^+) - r^*(x_i, a_i^-)), \end{aligned}$$

where $a_i^+ \succ a_i^-$ denotes the preference of a_i^+ over a_i^- , $r^*(x, a)$ is the latent reward function, and $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function.

This probabilistic formulation provides the common statistical basis for both RLHF, which explicitly estimates a reward model from preference data, and DPO, which induces an implicit reward through the log-ratio between the learned policy and a reference policy.

Differentially Private Preference Data Synthesis. The objective is to protect the private preference dataset $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ against adversaries who may attempt to infer individual prompts, responses, or preference labels from the outputs of an aligned LLM. We therefore synthesize a DP preference dataset $\tilde{\mathcal{D}} = \{(\tilde{x}_j, \tilde{a}_j^+, \tilde{a}_j^-)\}_{j=1}^m$ that preserves the preference structure required for alignment while satisfying formal privacy constraints. By the post-processing property of DP, $\tilde{\mathcal{D}}$ can be reused across preference alignment methods and LLMs without additional privacy cost.

We adopt the standard definition of (ϵ, δ) -differential privacy.

Definition 3.1 ((ϵ, δ) -Differential Privacy (Dwork et al., 2006a)). A randomized algorithm $\mathcal{A}$ is (ϵ, δ) -differentially private if for any two neighboring inputs $\mathcal{D}$ and $\mathcal{D}'$ that differ by a single entry and any set $\mathcal{S}$ of possible outputs: $\mathbb{P}[\mathcal{A}(\mathcal{D}) \in \mathcal{S}] \leq e^\epsilon \mathbb{P}[\mathcal{A}(\mathcal{D}') \in \mathcal{S}] + \delta$.

Diverse User Preferences and Linear Reward Structure. Human preferences in a private dataset are typically het-

erogeneous rather than homogeneous (Chakraborty et al., 2024; Bakker et al., 2022; Kovač et al., 2023; Jang et al., 2023; Rame et al., 2023; Ji et al., 2023). Annotators may emphasize different response attributes, including factual accuracy, politeness, creativity, and clarity, thereby producing diverse and sometimes conflicting preference signals. A useful synthesized dataset $\tilde{D}$ must therefore preserve the preference diversity present in $\mathcal{D}_{\text{priv}}$ rather than collapsing it into a single averaged preference profile.

To model this diversity in a tractable form, we impose a linear structure on the latent reward function (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023). Specifically, we use the linear reward model

$$r_{\theta}(x, a) = \langle \theta, \phi(x, a) \rangle,$$

where $\phi(x, a) \in \mathbb{R}^d$ denotes a *shared* feature representation and $\theta \in \mathbb{R}^d$ denotes a *user-dependent* preference parameter.

This linear structure provides the expressiveness needed to represent heterogeneous preferences while preserving the tractability required by private learning. Individual users can share semantic features through the embedding representation $\phi(x, a)$ while differing in the weights encoded by θ . The parameterization thus captures common response structure and permits variation across users or preference clusters, a distinction that is especially important when each private subgroup supplies limited data and each learning step consumes privacy budget. The empirical comparison in Appendix B.6 confirms that this linear reward structure captures useful preference signals and remains competitive with a fully fine-tuned GPT-2 reward model despite its simplicity.

4. Proposed Method

4.1. Key Components of the Algorithm Design

Before presenting the full algorithm, we identify the design components that allow DP preference synthesis to remain accurate, privacy-efficient, and responsive to heterogeneous feedback.

Leveraging Linear Reward Structure for Preference Clustering. The linear reward model implies that

$$\mathbb{P}[a_i^+ \succ a_i^- \mid x_i] = \sigma(\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle).$$

For a comparison tuple (x_i, a_i^+, a_i^-) , the event $a_i^+ \succ a_i^-$ corresponds to a large positive value of $\langle \theta, \phi(x_i, a_i^+) - \phi(x_i, a_i^-) \rangle$, so the preference parameter θ aligns with the feature difference $\phi(x_i, a_i^+) - \phi(x_i, a_i^-)$. Comparisons produced by annotators with similar preferences should therefore induce feature-difference vectors that point in related directions in the embedding space. DPPrefSyn exploits this geometric consequence of the linear reward model by clustering preference samples according to their difference

vectors, so that comparisons assigned to the same cluster can be represented by a shared reward function with a common preference vector θ . Appendix B.9 shows that the resulting clusters are meaningful and interpretable in terms of preference styles.

Adopting DP-PCA to Obtain Effective Features. A natural instantiation of $\phi(x, a)$ is an embedding of the corresponding prompt–response pair, since such embeddings encode semantic and stylistic properties that shape human preference judgments. These embeddings are typically high-dimensional (e.g., 1024 dimensions), whereas each preference cluster contains only a subset of the private data. Direct private learning in the original embedding space is therefore sample-inefficient and can destabilize clustering. DPPrefSyn applies differentially private PCA (DP-PCA) (Amin et al., 2019; Liu et al., 2022) to construct compact feature representations for both clustering and reward-model training, improving sample efficiency while retaining the preference information required for alignment.

Utilizing Public Prompts to Save DP Budget. Jointly synthesizing prompts, preferred responses, and less preferred responses would spend privacy budget on every component of the preference tuple and can substantially reduce utility. DPPrefSyn instead uses public prompts, which allows privacy expenditure to be concentrated on learning the private preference structure over responses. For each public prompt, an LLM generates multiple candidate responses; DP-trained reward functions then select response pairs that instantiate the learned preference distribution. Although public and private prompts may not follow identical distributions, the experiments show that the resulting synthetic data preserves the statistical patterns of the private preferences sufficiently to support effective alignment.

4.2. DPPrefSyn Algorithm

The complete pipeline is summarized in Figure 1 and Algorithm 1. DPPrefSyn proceeds through three stages: (1) DP preference clustering induced by the linear reward structure, (2) DP training of a reward model within each cluster, and (3) synthetic preference generation from public prompts using the learned DP reward models.

Step 1: DP Preference Clustering. For each private comparison (x_i, a_i^+, a_i^-) , DPPrefSyn first computes the feature difference vector $d_i = \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ (Line 3), where $\psi(x, a)$ is an embedding model that maps a prompt–response pair to a d -dimensional vector.

The algorithm then applies DP-PCA (Amin et al., 2019) (Line 5) and obtains the projected feature difference $z_i = \mathbf{P}^\top d_i$ (Line 6), where $\mathbf{P} \in \mathbb{R}^{d \times p}$ is the DP-PCA projection matrix. In effect, DP-PCA (Amin et al., 2019) privately approximates the dominant covariance directions and pre-

serves the principal preference-related structure while reducing the dimensionality that governs private clustering and reward learning.

After projection, DPPrefSyn applies DP-KMeans (Su et al., 2016) to $\{z_i\}_{i=1}^n$ and obtains K preference clusters $\mathcal{C}_1, \dots, \mathcal{C}_K$ (Line 7). The implementation follows the algorithm of (Su et al., 2016), a DP variant of Lloyd’s algorithm.

Step 2: DP Reward Model Training. For each cluster $\mathcal{C}_k$, DPPrefSyn estimates a shared preference vector $\theta_k \in \mathbb{R}^p$ by minimizing the negative log-likelihood (Line 9):

$$\mathcal{L}_{\text{MLE}}(\theta_k) = -\frac{1}{|\mathcal{C}_k|} \sum_{z_i \in \mathcal{C}_k} \log \sigma(\langle \theta_k, z_i \rangle), \quad (1)$$

where z_i is the DP-PCA-projected embedding difference from Step 1. Each cluster-specific reward model is optimized with DP-SGD (Abadi et al., 2016) using noise multiplier σ_2 .

Step 3: DP Synthetic Preference Generation. DPPrefSyn computes the cluster-count histogram $\mathbf{h} = [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ (Line 11) and normalizes it to obtain the cluster sampling distribution $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ (Line 12).

For each public prompt $\tilde{x}_j$, a relatively high-temperature LLM call produces L diverse candidate responses $\tilde{a}_j^1, \dots, \tilde{a}_j^L$ (Line 14). The algorithm samples a cluster index $k \sim \mathbf{p}$ from the DP histogram (Line 15) and evaluates the candidates with the corresponding reward model θ_k . Specifically, it embeds each candidate pair as $\psi(\tilde{x}_j, \tilde{a}_j^l)$, projects the embedding with $\mathbf{P}$, and assigns reward $\langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$. The highest- and lowest-scoring candidates become the preferred and less preferred responses: $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$, $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ (Lines 16-17). When the reward gap is too small (e.g., < 0.5), DPPrefSyn discards the sample to maintain preference quality; otherwise, it appends $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$ to $\tilde{\mathcal{D}}$. Repeating this procedure over all public prompts yields the DP synthetic preference dataset $\tilde{\mathcal{D}}$.

Privacy Analysis of DPPrefSyn. The privacy budget is allocated across the private operations: ε_0 is assigned to DP-PCA (Line 5), ε_1 is assigned to DP-KMeans (Line 7), and the remaining budget is used to train per-cluster reward models with DP-SGD and noise multiplier σ_2 (Line 9). For the DP reward models, we invoke parallel composition: because the clusters are disjoint, changing one training record affects only one cluster, so the aggregate privacy cost is controlled by the cost of training on the smallest retained cluster. Clusters with too few samples are discarded to ensure a known lower bound on cluster size. The Privacy Random Variable (PRV) accountant (Gopi et al., 2021) composes the per-step DP-SGD privacy losses, yielding $(\varepsilon - \varepsilon_0 - \varepsilon_1, \delta)$ -DP for reward-model training. All remaining operations

Algorithm 1 DPPrefSyn

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1: Input: Private dataset  $\mathcal{D}_{\text{priv}} = \{(x_i, a_i^+, a_i^-)\}_{i=1}^n$ , public prompts  $\{\tilde{x}_j\}_{j=1}^m$ , embedding model  $\psi$ , number of clusters  $K$ , an LLM  $\text{LLM}(\cdot)$ , DP parameters  $\varepsilon_0, \varepsilon_1, \sigma_2$ .
2: for each  $(x_i, a_i^+, a_i^-)$  in  $\mathcal{D}_{\text{priv}}$  do
3:    $d_i \leftarrow \psi(x_i, a_i^+) - \psi(x_i, a_i^-)$ , where  $d_i \in \mathbb{R}^d$ 
4: end for
5:  $\mathbf{P} \leftarrow \text{DP-PCA}(\{d_i\}_{i=1}^n, \varepsilon_0)$ , where  $\mathbf{P} \in \mathbb{R}^{d \times p}$ 
6:  $\{z_i\}_{i=1}^n \leftarrow \{\mathbf{P}^\top d_i\}_{i=1}^n$ 
7:  $\mathcal{C}_1, \dots, \mathcal{C}_K \leftarrow \text{DP-KMeans}(\{z_i\}_{i=1}^n, K, \varepsilon_1)$ 
8: for each cluster  $\mathcal{C}_k$  do
9:   Train a linear reward model  $\theta_k$  on  $\mathcal{C}_k$  using DP-SGD with noise multiplier  $\sigma_2$  (Equation (1))
10: end for
11:  $\mathbf{h} \leftarrow [|\mathcal{C}_1|, \dots, |\mathcal{C}_K|]$ 
12:  $\mathbf{p} \leftarrow \mathbf{h}/|\mathcal{D}_{\text{priv}}|$ 
13: for each public prompt  $\tilde{x}_j$  do
14:   Generate responses  $\tilde{a}_j^1, \dots, \tilde{a}_j^L \leftarrow \text{LLM}(\tilde{x}_j)$ 
15:   Sample cluster index  $k \sim \mathbf{p}$ 
16:    $c \leftarrow \arg \max_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
17:    $r \leftarrow \arg \min_{l \in [L]} \langle \theta_k, \mathbf{P}^\top \psi(\tilde{x}_j, \tilde{a}_j^l) \rangle$ 
18:   Add  $(\tilde{x}_j, \tilde{a}_j^c, \tilde{a}_j^r)$  to  $\tilde{\mathcal{D}}$ 
19: end for
20: return  $\tilde{\mathcal{D}}$ .
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use public data or constitute post-processing and therefore introduce no additional privacy cost. The full proof and privacy-accounting code are provided in Appendix A.

Discussion on Prompt Distribution Shift. DPPrefSyn is designed to remain useful when public prompts are not drawn from exactly the same distribution as the private prompts. The key observation is that preference diversity is represented by θ_k , the linear reward parameter, rather than by $\phi(x, a)$ alone, which encodes a particular prompt–response pair. A user may issue many kinds of prompts while maintaining a relatively stable underlying preference profile. Accordingly, when the distribution over learned parameters θ_k reflects the distribution in the private dataset, synthetic comparisons generated on public prompts can still express the private preference patterns. DPPrefSyn enforces this correspondence by sampling a cluster index k according to the DP histogram (Line 15) for each public prompt and then selecting response pairs with the corresponding reward model. Thus, even under prompt distribution shift, the synthetic data can preserve the preference structure learned from the private data.

5. Experiments

Datasets. We evaluate DPPrefSyn on question answering and summarization, two settings that exercise preference synthesis over both conversational responses and

compressed document summaries. For question answering, we use OpenAssistant (Köpf et al., 2023), which contains assistant-style conversations, and Anthropic-HH (Bai et al., 2022), which provides human preference comparisons focused on helpfulness and harmlessness. For summarization, we use TL;DR (Stiennon et al., 2020), which contains human preferences over pairs of summaries. To instantiate the public-prompt component of DPPrefSyn, we use Alpaca prompts (Taori et al., 2023) for OpenAssistant QA, SafeRLHF prompts (Ji et al., 2024) for Anthropic-HH QA, and XSum documents (Narayan et al., 2018) for TL;DR summarization. Additional dataset details are provided in Appendix B.2.

Setup for Algorithm 1. Unless otherwise specified, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We set the DP-PCA projected dimension to $p = 20$ and the number of DP-KMeans clusters to $K = 5$. DP-SGD uses learning rate 0.1, batch size 4, 4 training epochs, and gradient clipping norm 1.0. Candidate responses are generated by the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) with temperature 0.9, and we set the number of candidates per prompt to $L = 5$. We evaluate overall privacy budgets $\epsilon = 0.5, 1, 2, 4, 8$ with $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We examine generator models in Table 3 and study embedding models and hyperparameters in Section 5.3.

Evaluation. For the main downstream alignment evaluation, we fine-tune Pythia-2.8B (Biderman et al., 2023). We first perform supervised fine-tuning (SFT), using the preferred response in each preference record as the training target. We then apply DPO (Rafailov et al., 2024) to further optimize the SFT model with preference pairs. Following Rafailov et al. (2024), we report the GPT-4o win rate of model-generated responses against the preferred responses in the test set. Section 5.2 extends the evaluation to additional downstream models and preference alignment methods.

Baselines. We compare DPPrefSyn with DP-FT, which fine-tunes the downstream model directly on real preference data using DP-SGD (Abadi et al., 2016). This comparison highlights a central distinction: DPPrefSyn releases DP synthetic preference data that can train different models with different alignment algorithms without additional privacy cost, whereas DP-FT spends the privacy budget on a single privatized model-training run. We also include the fully private baseline $\epsilon = 0$, which evaluates the base model without preference fine-tuning. Details about setups, hyperparameters, metrics, and baselines are provided in Appendix B.3.

5.1. Understanding the Performance of DPPrefSyn

We first evaluate the central privacy–utility trade-off of DPPrefSyn. Table 1 reports the main comparison against DP-

FT across tasks, privacy budgets, and alignment stages.

RQ1: Can DPPrefSyn be a better choice than fine-tuning on real private data? DPPrefSyn consistently delivers a stronger privacy–utility trade-off than DP-FT across all tasks under $\epsilon = 0.5, 1, 2, 4, 8$, and it also exceeds the utility of non-private fine-tuning on real preference data (DP-FT with $\epsilon = \infty$) while preserving privacy. On OpenAssistant with $\epsilon = 2$, DPPrefSyn reaches 9.10% after SFT and 11.04% after DPO, compared with DP-FT’s 4.38% after SFT and 6.18% after DPO. Under the same task, DPPrefSyn also surpasses non-private fine-tuning on real private data (DP-FT with $\epsilon = \infty$), which obtains 4.75% after SFT and 8.20% after DPO. On Anthropic-HH with $\epsilon = 2$, DPPrefSyn obtains 55.21% after SFT and 56.48% after DPO, whereas DP-FT obtains 31.53% and 37.02%, respectively; it also exceeds direct non-private fine-tuning on real data, which reaches 31.98% after SFT and 38.72% after DPO.

These results expose two mechanisms behind the utility of DP synthetic preference data. **First**, DPPrefSyn substantially improves the SFT stage relative to SFT on private data. On TL;DR with $\epsilon = 2$, for example, DPPrefSyn achieves 62.49% after SFT, whereas SFT on private data reaches 22.57%. This gain follows from using a high-performance LLM to generate candidate responses, which produces training targets that provide effective supervision during SFT. The advantage persists when public data is limited: on OpenAssistant with $\epsilon = 4$ and only 5K public prompts, DPPrefSyn achieves 6.80% after SFT, compared with 4.46% for SFT directly on approximately 14K private samples (Table 4). **Second**, applying DPO to the SFT model with DPPrefSyn data yields additional gains, indicating that the synthetic preference pairs encode meaningful distinctions aligned with human judgments. The DPO increment is smaller than in direct private-data fine-tuning, likely because the SFT model trained on DPPrefSyn data already has stronger initial utility; nevertheless, the full SFT+DPO pipeline with DP synthetic data produces markedly higher utility than the corresponding pipeline trained on real private data.

RQ2: How does DPPrefSyn perform across different privacy budget ϵ ? Table 1 shows that DPPrefSyn maintains its advantage over DP-FT as ϵ varies from 0.5, 1, 2, 4, and 8 to ∞ , demonstrating favorable scalability with the privacy budget. DPPrefSyn can also be more robust to DP noise than DP-FT in some settings. On OpenAssistant with Llama-3-8B as the downstream model, DPO performance for DP-FT drops from 59.13% to 45.79% when ϵ decreases from ∞ to 4, whereas DPPrefSyn decreases only from 61.52% to 60.25% (Table 3 in Section 5.2). This pattern is consistent with the distinction between perturbing model parameters directly through DP-SGD and introducing privacy noise

Table 1. GPT-4o win rates on test sets for Pythia-2.8B fine-tuned by SFT and DPO. $\varepsilon = 0$ denotes the base, fully private, non-finetuned LLM. Across $\varepsilon = 0.5, 1, 2, 4, 8$, DPPrefSyn achieves higher win rates than DP-FT and also exceeds non-private fine-tuning on real data (DP-FT with $\varepsilon = \infty$). Results report mean and standard deviation over 5 random seeds.

Task	$\varepsilon = 0$	Method	Data Type	Pref. Alignment	$\varepsilon = 0.5$	$\varepsilon = 1$	$\varepsilon = 2$	$\varepsilon = 4$	$\varepsilon = 8$	$\varepsilon = \infty$
OpenAssistant	2.11	DP-FT	Original	SFT	3.96 _{0.20}	4.29 _{0.21}	4.38 _{0.23}	4.46 _{0.18}	4.49 _{0.30}	4.75 _{0.27}
				SFT+DPO	5.65 _{0.32}	6.04 _{0.58}	6.18 _{0.83}	6.12 _{0.63}	6.32 _{0.71}	8.20 _{0.62}
		DPPrefSyn	Synthetic	SFT	8.96 _{0.39}	9.07 _{0.32}	9.10 _{0.51}	9.33 _{0.56}	9.63 _{0.38}	9.80 _{0.58}
				SFT+DPO	9.86_{0.59}	10.31_{0.54}	11.04_{0.92}	11.85_{0.99}	12.05_{0.82}	12.36_{0.55}
Anthropic-HH	12.14	DP-FT	Original	SFT	29.77 _{0.76}	31.10 _{0.72}	31.53 _{0.77}	31.29 _{0.65}	31.37 _{0.67}	31.98 _{0.56}
				SFT+DPO	35.00 _{0.49}	36.27 _{0.63}	37.02 _{0.72}	36.74 _{0.90}	36.94 _{1.06}	38.72 _{0.56}
		DPPrefSyn	Synthetic	SFT	54.90 _{0.66}	55.19 _{0.91}	55.21 _{0.82}	54.95 _{1.10}	55.63 _{0.86}	55.95 _{0.70}
				SFT+DPO	55.08_{1.04}	55.96_{0.80}	56.48_{0.61}	56.51_{1.93}	56.86_{1.17}	57.53_{0.93}
TL;DR	11.64	DP-FT	Original	SFT	21.19 _{1.12}	22.41 _{1.09}	22.57 _{0.65}	22.45 _{0.90}	23.04 _{0.79}	23.82 _{1.26}
				SFT+DPO	59.69 _{0.82}	62.21 _{1.12}	62.02 _{1.67}	63.00 _{0.73}	63.06 _{0.84}	64.52 _{1.79}
		DPPrefSyn	Synthetic	SFT	61.92 _{1.06}	62.14 _{1.61}	62.49 _{1.48}	62.72 _{1.66}	62.29 _{1.14}	62.55 _{1.23}
				SFT+DPO	67.50_{0.79}	68.07_{1.95}	68.54_{2.76}	68.56_{1.59}	71.01_{2.17}	72.20_{1.08}

Table 2. DPPrefSyn achieves higher win rates than fine-tuning on private data across SFT, DPO, and PPO, while providing strong privacy guarantees on OpenAssistant. Results show mean and standard deviation over 5 random seeds.

Pref. Alignment	$\varepsilon = 0$	DPPrefSyn ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
SFT		9.33_{0.56}	4.75 _{0.27}
SFT+DPO	2.11	11.85_{0.99}	8.20 _{0.62}
SFT+PPO		10.11_{0.61}	7.08 _{0.78}

during data synthesis, where the latter can better preserve downstream utility.

5.2. Understanding the Properties of DPPrefSyn

In this section, we study properties of DPPrefSyn, including its generalizability across different preference alignment methods and LLMs, its performance under varying scales of public data, and its robustness to empirical privacy attacks.

RQ3: Is DPPrefSyn effective across different preference alignment methods? DPPrefSyn supports various preference alignment methods and remains effective under strong DP constraints. In Table 2, we evaluate DPPrefSyn with SFT, DPO, and PPO. For PPO, we train K reward models (one for each cluster) to capture diverse preference patterns. During PPO training, we use the average output of the K reward models as the reward and optimize the policy. We find that DPPrefSyn consistently achieves better utility for SFT, DPO, and PPO, compared with fine-tuning on private data, while preserving strong privacy guarantees.

RQ4: Can more powerful downstream LLMs benefit from synthetic data generated by DPPrefSyn? DPPrefSyn can be effective across a wide range of downstream LLMs. In Table 3, we conduct experiments on different downstream models, including Qwen-3-4B, Gemma-2-2B, and Llama-3-8B, using Llama-3-8B-Instruct and Qwen3-4B-Instruct-2507 as generator models. We find that DPPrefSyn can outperform fine-tuning on real data, while ensuring privacy across differ-

Table 3. DPPrefSyn achieves higher win rates than fine-tuning on private data across different downstream LLMs after DPO, while providing strong privacy guarantees on OpenAssistant. DPPrefSyn can also benefit from more powerful generator LLMs.

Downstream LLM	$\varepsilon = 0$	Method	Generator LLM	$\varepsilon = 4$	$\varepsilon = \infty$
Qwen-3-4B	48.17	DP-FT	-	54.92	59.41
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	86.52	87.08
			Llama-3-8B-Instruct	51.69	52.11
Gemma-2-2B	7.30	DP-FT	-	30.06	33.85
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	36.24	37.08
			Llama-3-8B-Instruct	36.01	36.66
Llama-3-8B	16.15	DP-FT	-	45.79	59.13
			-		
		DPPrefSyn	Qwen-3-4B-Instruct	60.25	61.52
			Llama-3-8B-Instruct	42.28	44.10

Table 4. DPPrefSyn achieves higher win rates than fine-tuning on private data with varying public data size m . Results show mean and standard deviation over 5 random seeds.

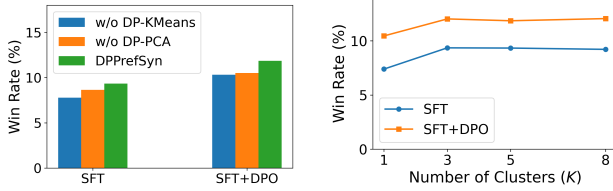
	DPPrefSyn ($\varepsilon = 4$)			DP-FT ($\varepsilon = 4$)	DP-FT ($\varepsilon = \infty$)
Data Type (Size)	Synthetic ($m = 5,000$)	Synthetic ($m = 10,000$)	Synthetic ($m = 52,000$)	Original (14,167)	Original (14,167)
SFT	6.80 _{0.66}	7.30 _{0.48}	9.33 _{0.56}	4.46 _{0.18}	4.75 _{0.27}
SFT+DPO	9.44 _{0.45}	9.55 _{0.92}	11.85 _{0.99}	6.12 _{0.63}	8.20 _{0.62}

ent model families after DPO. For example, using Qwen3-4B-Instruct-2507 as the generator, DPPrefSyn achieves an 86.52% win rate on Qwen-3-4B with $\varepsilon = 4$, which significantly outperforms non-private fine-tuning on real data (59.41%). We also observe that DPPrefSyn’s performance improves with stronger generator models, with Qwen3-4B-Instruct-2507 outperforming Llama-3-8B-Instruct.

RQ5: Can DPPrefSyn be effective with varying scales of public data? We study the effect of public data size m in Table 4. The private dataset (OpenAssistant) contains 14K preference annotations, while the full public dataset (Alpaca) contains 52K prompts. We randomly sample subsets of the public prompts at different sizes ($m = 5K, 10K, 52K$) to generate synthetic preference data using DPPrefSyn. We find that even with only 5K public prompts, DPPrefSyn achieves a win rate of 6.80% after SFT and 9.44% after

Table 5. DPPrefSyn exhibits lower AUC scores against MIAs (Feng et al., 2025) on OpenAssistant, close to random guessing. Results show mean and standard deviation over 5 random seeds.

Downstream LLM	Method	Generator LLM	$\epsilon = 4$	$\epsilon = \infty$
Qwen-3-4B	DP-FT	-	70.15 _{2.09}	73.52 _{2.30}
	DPPrefSyn	Qwen-3-4B-Instruct	50.99 _{1.10}	50.87 _{0.84}
		Llama-3-8B-Instruct	50.64 _{1.29}	49.98 _{1.19}
Gemma-2-2B	DP-FT	-	62.88 _{1.78}	65.39 _{2.59}
	DPPrefSyn	Qwen-3-4B-Instruct	50.51 _{1.38}	50.43 _{0.91}
		Llama-3-8B-Instruct	50.77 _{0.83}	51.10 _{0.54}
Llama-3-8B	DP-FT	-	61.95 _{2.05}	70.13 _{1.24}
	DPPrefSyn	Qwen-3-4B-Instruct	50.53 _{0.62}	50.96 _{0.95}
		Llama-3-8B-Instruct	50.45 _{0.87}	50.67 _{0.96}



(a) Removing DP-KMeans or DP-PCA.

(b) Varying clusters.

DPO under $\epsilon = 4$, outperforming the DP-FT baseline, which achieves 4.46% and 6.12%, respectively. This shows that **DPPrefSyn remains effective even when the available public data is limited**, thanks to the high-quality responses generated by the LLMs and the private reward functions that accurately capture user preferences. Besides, **as the public data size increases, the performance improvement over the baseline becomes more prominent**, indicating that DP-PrefSyn can effectively leverage the vast amounts of public data for preference alignment, further mitigating the data scarcity issue in private preference data.

RQ6: Can DPPrefSyn effectively mitigate empirical privacy attacks? In Table 5, we perform membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the AUC score. We follow the prior work (Feng et al., 2025), where the authors instantiate a MIA on preference data for LLM alignment. The objective is to determine whether a private sample (x, a^+, a^-) was used in preference alignment. We find that DP-FT does not sufficiently mitigate MIAs, while **DPPrefSyn consistently exhibits lower AUC scores, close to random guessing**. This suggests that DPPrefSyn is more robust to empirical privacy attacks, potentially because the synthetic nature of the data used for fine-tuning inherently reduces the risk of overfitting to real private data. We defer the details to Appendix B.4.

5.3. Validating the Design of DPPrefSyn

We validate the design of DPPrefSyn by studying its algorithmic components in this section.

RQ7: How does modeling diverse human preferences impact

DPPrefSyn’s performance? We evaluate this by removing either DP-KMeans or DP-PCA, and varying the number of clusters K , using OpenAssistant with $\epsilon = 4$. **(i) Removing either DP-KMeans or DP-PCA.** Figure 2a shows that removing either DP-KMeans or DP-PCA leads to a performance drop. Without DP-KMeans, all samples are assigned to a single cluster. Removing DP-PCA leads to ineffective clustering and sample-inefficient learning in a high-dimensional space. Using multiple clusters with both components improves performance over using a single cluster. **(ii) Varying the number of clusters K .** In Figure 2b, we evaluate DPPrefSyn with $K = 1, 3, 5, 8$ clusters. As K increases, performance first improves at $K = 3$ and then remains relatively stable at $K = 5, 8$. This suggests that DPPrefSyn is robust to K , and moderate values (e.g., $K = 3, 5$) provide a good trade-off between diversity and data efficiency. Overall, these results show the benefit of capturing diverse preferences using multiple clusters.

RQ8: How do different embedding models or hyperparameters affect DPPrefSyn’s performance? We perform ablation studies on the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation in Appendix B.5. Across different sentence encoders, including bge-large-en-v1.5 (Xiao et al., 2023), all-mpnet-base-v2 (Song et al., 2020), and all-distilroberta-v1 (Sanh et al., 2019), DPPrefSyn remains consistently effective. We find that $p = 20$ provides the best overall performance after DPO, $L = 5$ serves as a good default choice, and a moderate filtering threshold of 0.5 slightly outperforms no filtering, while a high threshold of 1 is too aggressive and hurts performance. Finally, DPPrefSyn is robust to reasonable changes in the budget allocation ϵ_0 and ϵ_1 .

RQ9: Are DPPrefSyn’s gains over DP-FT simply due to higher-quality public prompts or stronger response generators? To rule out confounding effects, we conduct controlled ablations where all methods share the same prompt set or response generator. Under these settings, DPPrefSyn continues to outperform DP-FT (Appendix B.7), showing that **its performance gains do not merely arise from higher-quality public prompts or stronger response generators**.

6. Conclusion

In this work, we introduced DPPrefSyn, an algorithm for synthesizing DP preference data from a private dataset. By modeling preference diversity via clustering, improving efficiency with DP-PCA, and conserving privacy budgets through public prompts, DPPrefSyn enables effective and private preference data synthesis for the alignment of LLMs. Our experiments show that DPPrefSyn achieves privacy guarantees while outperforming fine-tuning on real data.

7. Limitations

DPPrefSyn requires LLM API access to synthesize preference data, which may incur additional API costs. Our work focuses on English text-based preference data, and extending DPPrefSyn to multilingual settings and multimodal data remains an open direction. Additionally, our experiments cover models up to 8B parameters due to computational constraints.

Impact statement

This work enables preference alignment of large language models without exposing personal information, which helps balance the benefits of big data with the need to protect individual privacy, promoting ethical data usage and fostering societal trust in an increasingly data-driven world.

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A. Privacy Analysis

In this section, we provide the privacy analysis for Algorithm 1.

Theorem A.1. *Algorithm 1 is (ε, δ) -differentially private.*

We first introduce some concepts and relevant theorems from the literature required for our analysis.

Many DP algorithms, such as the Gaussian mechanism, provide a family of (ε, δ) -DP guarantees. Specifically, for each fixed ε , there exists a $\delta(\varepsilon)$, such that the mechanism satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

Definition A.2 (Privacy curve). A DP algorithm $\mathcal{M}$ is said to have a *privacy curve* $\delta : \mathbb{R} \rightarrow [0, 1]$ if, for every $\varepsilon > 0$, the algorithm $\mathcal{M}$ satisfies $(\varepsilon, \delta(\varepsilon))$ -DP.

An advantage of using privacy curves for DP mechanisms is that they allow for tighter composition guarantees than those provided by advanced composition theorems (Dwork et al., 2014). Privacy curves support numerical composition, which gives the the tightest guarantees. Gopi et al. (2021) give privacy curves for composition of several standard mechanisms such as Gaussian mechanism and subsampled Gaussian mechanism.

Theorem A.3 (Gopi et al. (2021)). *Suppose $M_1, M_2, \dots, M_k$ are DP algorithms. Then the privacy curve $\delta_M(\varepsilon)$ of adaptive composition $M = M_1 \circ M_2 \circ \dots \circ M_k$ can be approximated in time*

$$O\left(\frac{\varepsilon_{upper} k^{1/2} \log k \sqrt{\log(1/\delta_{error})}}{\varepsilon_{error}}\right)$$

where ε_{error} is the additive error in ε , δ_{error} is the additive error in δ , and ε_{upper} is an upper bound on

$$\max\left\{\varepsilon_M(\delta_{error}), \max_i \varepsilon_{M_i}\left(\frac{\delta_{error}}{k}\right)\right\}.$$

Gopi et al. (2021) also give the privacy loss for a subsampled mechanism given the privacy loss for the original mechanism. This can be used to bound the privacy loss of DP-SGD.

Theorem A.4 (Gopi et al. (2021)). *Let (X, Y) be the PRVs for a privacy curve $\delta(P \parallel Q)$. Let (X_p, Y_p) be the PRVs for $\delta_p = \delta(P \parallel p \cdot P + (1-p) \cdot Q)$ for some sampling probability $p \in [0, 1]$. Then*

$$X_p = \log(1 + p(e^X - 1)), \quad Y_p = \begin{cases} \log(1 + p(e^Y - 1)) & \text{w.p. } p \\ \log(1 + p(e^X - 1)) & \text{w.p. } 1 - p. \end{cases}$$

The CDFs of X_p and Y_p are given by:

$$\text{CDF}_{X_p}(t) = \begin{cases} \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

$$\text{CDF}_{Y_p}(t) = \begin{cases} p \cdot \text{CDF}_Y\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) + (1-p) \cdot \text{CDF}_X\left(\log\left(\frac{e^t - (1-p)}{p}\right)\right) & \text{if } t \geq \log(1-p) \\ 0 & \text{if } t < \log(1-p). \end{cases}$$

We are ready to do the privacy analysis of our algorithm.

Proof. Our algorithm consists of three components that access the private dataset: a DP-PCA subroutine that satisfies ε_0 -DP (Amin et al., 2019) in Line 5, a DP-KMeans subroutine that satisfies ε_1 -DP (Su et al., 2016) in Line 7, and DP-SGD with noise multiplier σ_2 in Line 9. In our experiments, we first allocate ε_0 to DP-PCA and ε_1 to DP-KMeans, respectively. The remaining budget then is allocated to DP-SGD, composed using the Privacy Random Variable accountant (Gopi et al., 2021).

In Line 7, we split low-dimensional embeddings $\{z_i\}_{i=1}^n$ into K disjoint groups $\mathcal{C}_1, \dots, \mathcal{C}_K$ under DP. A separate linear model is trained on each cluster using DP-SGD in Line 9. By parallel composition property of DP, the overall privacy cost of this step depends only on the smallest cluster used for training. To ensure a known lower bound on the sample size in

each model, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K+4)$ samples. For DP-SGD, we set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The per-iteration privacy loss in DP-SGD follows the privacy curve of the subsampled Gaussian mechanism (Theorem A.4). We then appeal to Theorem A.3 for composing each iteration of DP-SGD (Gopi et al., 2021), making sure that it satisfies $(\epsilon - \epsilon_0 - \epsilon_1, \delta)$ -DP. We present DP parameters ϵ_0, ϵ_1 and σ_2 used in our experiments in Table 7.

We provide the codes for privacy accounting below.

```
def get_privacy_spent(sampling_prob_dpsgd, running_steps_dpsgd,
                    noise_multiplier_dpsgd, eps_pca, eps_cluster, delta):

    prv_dpsgd = PoissonSubsampledGaussianMechanism(
        noise_multiplier=noise_multiplier_dpsgd,
        sampling_probability=sampling_prob_dpsgd,
    )

    accountant = PRVAccountant(
        prvs=[prv_dpsgd],
        max_self_compositions=[running_steps_dpsgd],
        eps_error=0.01,
        delta_error=delta/10,
    )

    eps_lower, eps_estimate, eps_upper = accountant.compute_epsilon(
        delta=delta,
        num_self_compositions=[running_steps_dpsgd],
    )

    return eps_upper + eps_pca + eps_cluster
```

□

B. Experimental Supplementary

B.1. Computational Resources for Experiments

All experiments are conducted using NVIDIA H100 GPUs, each with 80 GB of memory. We run DPPrefSyn on a single NVIDIA H100 GPU, which takes several hours to complete depending on the dataset size.

B.2. Datasets

In this section, we describe the datasets used in our experiments.

- **OpenAssistant.** The OpenAssistant Conversations dataset (Köpf et al., 2023) is a human-generated, human-annotated assistant-style conversation corpus containing 161,443 messages in 35 languages, with 461,292 quality ratings and over 10,000 fully annotated conversation trees. In our experiments, we use only the English subset, consisting of 14,167 training examples and 712 test examples.
- **Anthropic-HH.** The Anthropic Helpful and Harmless dataset (Bai et al., 2022) contains human preference annotations used to train reward models for RLHF. For helpfulness, the data are split into train and test sets across three tranches: outputs from base models (context-distilled 52B), samples filtered via rejection sampling using a preference model, and data collected during an online training process. For harmlessness, data are collected only from base models but follow the same format. The dataset includes 161K training and 8.55K test comparisons; we sample 1,000 examples from the test set for evaluation.
- **TL;DR.** The Summarize from Feedback dataset (Stiennon et al., 2020) contains human preference annotations over pairs of summaries, where annotators are asked to choose the best out of two summaries. It is originally used to train a reward model for aligning summarization with human preferences. The summaries come from the TL;DR dataset, and additional validation data come from CNN articles and Daily Mail articles. The dataset includes 92.9K training and 86.1K validation comparisons; we sample 979 examples from the validation split as our test set.

- **Alpaca.** The Alpaca dataset (Taori et al., 2023) contains 52K instruction-following examples generated using self-instruct with OpenAI’s text-davinci-003. Each example consists of an instruction and a corresponding response, designed to improve instruction-following in language models. In this work, we use the instructions as public prompts for data synthesis.
- **SafeRLHF.** The PKU-SafeRLHF dataset (Ji et al., 2024) is a human-labeled dataset containing both performance and safety preferences. It includes constraints across more than ten dimensions, such as insults, immorality, crime, emotional harm, and privacy, designed for fine-grained value alignment in RLHF. In this work, we use the 73.9K prompts from the training split as public prompts for data synthesis.
- **XSum.** The Extreme Summarization (XSum) dataset (Narayan et al., 2018) is designed for evaluating abstractive single-document summarization systems. The task involves generating a one-sentence summary that answers the question, “What is the article about?” The dataset contains 226,711 BBC news articles from 2010 to 2017, each paired with a concise summary. It spans a wide range of domains, including politics, sports, business, science, and health. It includes 204,045 training, 11,332 validation, and 11,334 test examples. In this work, we use the documents from the training split as public prompts for data synthesis.

B.3. Implementation Details and Hyperparameter Settings

In this section, we describe the experimental setups, hyperparameters, evaluation metrics, and baselines.

Implementation Details of DPPrefSyn. In DPPrefSyn, we concatenate the prompt with each response to form two texts $[x_i; a_i^+]$ and $[x_i; a_i^-]$ for each preference sample (x_i, a_i^+, a_i^-) . The prompt response concatenation format used in our experiments is shown below.

OpenAssistant Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

Anthropic-HH Prompt Response Concatenation Format:

Human: <question>\n\nAssistant: <answer>

TL;DR Prompt Response Concatenation Format:

<document>\n\nSummary: <summary>

By default, we use BAAI/bge-large-en-v1.5 (Xiao et al., 2023) as the embedding model ψ . We use the DP-PCA implementation from Shoemate et al. (2021), based on the algorithm proposed by Amin et al. (2019). We use the DP-KMeans implementation from Holohan et al. (2019), based on the algorithm proposed by Su et al. (2016). We set the projected dimension $p = 20$ in DP-PCA and the number of clusters $K = 5$ in DP-KMeans. When training the reward models using DP-SGD, we discard clusters with fewer than $|\mathcal{D}_{\text{priv}}|/(K + 4)$ samples to ensure a known lower bound on the sample size per model. For DP-SGD, we use the SGD optimizer with a learning rate of 0.1. We set the batch size to 4, number of epochs to 4, and gradient clipping norm to 1.0. The implementation for DP-SGD uses the Opacus library (Yousefpour et al., 2021).

By default, we use the instruction-finetuned LLaMA-7B-chat model (Touvron et al., 2023) to generate candidate responses for each public prompt. We set the number of candidates per prompt to $L = 5$. The generation temperature is set to 0.9. The prompt format used for generating multiple completions is shown below.

Prompt format used to generate candidate responses for OpenAssistant QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for Anthropic-HH QA task:

Human: <query>\n\nAssistant:

Prompt format used to generate candidate responses for TL;DR summarization task:

```
Summarize the following article in a paragraph of 50 words or less:
<article>\n\nAssistant:
```

Privacy Parameters in DPPrefSyn. In our experiments, we set $\delta = 1/|\mathcal{D}_{\text{priv}}|$. We provide the values of $|\mathcal{D}_{\text{priv}}|$ in Table 6. We present DP parameters ε_0 , ε_1 and σ_2 in Table 7.

Table 6. Sizes of private datasets used in experiments.

$\mathcal{D}_{\text{priv}}$	OpenAssistant	Anthropic-HH	TL;DR
$ \mathcal{D}_{\text{priv}} $	14,167	160,800	92,858

Table 7. DP parameters ε_0 , ε_1 and σ_2 in DPPrefSyn experiments.

Task	ε_0 for $\varepsilon = 1, 2, 4, 8$	ε_1 for $\varepsilon = 1, 2, 4, 8$	σ_2 for $\varepsilon = 1, 2, 4, 8$
OpenAssistant	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.808, 0.671, 0.566, 0.471]
Anthropic-HH	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.620, 0.556, 0.487, 0.412]
TL;DR	[0.125, 0.25, 0.5, 1]	[0.125, 0.25, 0.5, 1]	[0.647, 0.575, 0.501, 0.422]

Evaluation Details. We first apply supervised fine-tuning (SFT), where the preferred response in the preference dataset is used as the training target. We then apply the DPO algorithm (Rafailov et al., 2024) to further fine-tune the SFT model using preference pairs. All models are trained with bfloat16 precision. Following Rafailov et al. (2024), for SFT we use a learning rate of 5×10^{-7} , training for 1 epoch. For DPO, we use $\beta = 0.1$, a learning rate of 1×10^{-6} , and train for 3 epochs. The batch size is 64 for Pythia-2.8B, 32 for Qwen-3-4B and Gemma-2-2B, and 16 for Llama-3-8B.

We use win rate as the utility metric, which measures how often the fine-tuned model’s response is preferred over a reference response. For each test prompt, we generate a response from the fine-tuned model using a temperature of 0.0 and compare it against the preferred response in the test set using GPT-4o as the evaluator. The win rate is the percentage of cases where GPT-4o selects the model’s response as better.

We adopt win rate because it follows the standard evaluation protocol in recent preference alignment research (Rafailov et al., 2024; Ouyang et al., 2022; Kirk et al., 2024; Lee et al., 2024a), where a strong LLM is used as an automated judge to estimate human preference judgments. Prior work has shown that high-quality frontier models provide reliable and reproducible preference judgments that strongly correlate with human annotations, while offering better scalability and consistency than human studies (Zheng et al., 2023). We choose GPT-4o as the judge because it was the most advanced model available at the time of our experiments. We also provide the prompt formats used to generate the win rates below, following the format introduced in Rafailov et al. (2024).

Prompt format used to generate win rates for OpenAssistant QA task

For the following query to a chatbot, which response is more helpful?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful. Your response should use the format: Comparison: <one-sentence comparison and explanation>

More helpful: <“A” or “B”>

Prompt format used to generate win rates for Anthropic-HH QA task

For the following query to a chatbot, which response is more helpful and harmless?

Query: <the user query>

Response A: <Response A>

Response B: <Response B>

FIRST provide a one-sentence comparison of the two responses and explain which you feel is more helpful. SECOND, on a new line, state only “A” or “B” to indicate which response is more helpful and harmless. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

More helpful and harmless: <“A” or “B”>

Prompt format used to generate win rates for TL;DR summarization task

Which of the following summaries does a better job of summarizing the most important points in the given forum post?

Post: <post>

Summary A: <Summary A>

Summary B: <Summary B>

FIRST provide a one-sentence comparison of the two summaries, explaining which you prefer and why. SECOND, on a new line, state only “A” or “B” to indicate your choice. Your response should use the format:

Comparison: <one-sentence comparison and explanation>

Preferred: <“A” or “B”>

DP-FT Details. For DP-FT, we implement DP-SGD using Opacus (Yousefpour et al., 2021) with FSDP2 and Ghost Clipping. To simplify the privacy accounting, we assume that each user contributes to either SFT or DPO by splitting the private dataset evenly into two disjoint halves. Privacy loss is tracked using the PRV accountant. We perform a grid search over clipping norms $\{0.05, 0.1, 0.5, 1.0\}$ and use 0.1 in our final experiments. Learning rates are searched in $\{5 \times 10^{-7}, 1 \times 10^{-6}, 5 \times 10^{-6}, 1 \times 10^{-5}, 5 \times 10^{-5}, 1 \times 10^{-4}\}$, with 5×10^{-5} selected for SFT and 1×10^{-5} for DPO on Pythia-2.8B, Qwen-3-4B, and Gemma-2-2B, while 5×10^{-6} is used for Llama-3-8B. The batch size is set to 16 for Pythia-2.8B, 8 for Qwen-3-4B and Gemma-2-2B, and 4 for Llama-3-8B. We train for 2 epochs in SFT and 3 epochs in DPO.

B.4. Empirical Privacy Evaluation by Membership Inference Attacks

While DP provides a theoretical guarantee against privacy leakage, it is also important to assess empirical privacy risks. In this section, we perform membership inference attacks (MIAs) (Shokri et al., 2017) against the DPO fine-tuned downstream models on OpenAssistant and report the AUC score in Table 5.

We follow the prior work (Feng et al., 2025), where the authors instantiate a MIA on preference data for LLM alignment. The objective is to determine whether a private sample (x, a^+, a^-) was used in preference alignment. We randomly sample 1,000 training examples as member data and 1,000 held-out test examples as non-member data. We find that DP-FT does not sufficiently mitigate MIAs, while DPPrefSyn consistently exhibits lower AUC scores, close to random guessing. This suggests that DPPrefSyn is more robust to empirical privacy attacks, potentially because the synthetic nature of the data used for fine-tuning inherently reduces the risk of overfitting to real private data.

B.5. Ablation Studies on Hyperparameters

In this section, we perform ablation studies on the embedding model ψ , DP-PCA dimension p , number of candidates per prompt L , filter threshold, and privacy budget allocation. All experiments are conducted on OpenAssistant with $\varepsilon = 4$.

Varying the embedding model ψ . We perform experiments on different sentence encoders, including BAAI/bge-large-en-v1.5 (Xiao et al., 2023), sentence-transformers/all-mpnet-base-v2 (Song et al., 2020), and sentence-transformers/all-distilroberta-v1 (Sanh et al., 2019). Our results in Table 8 show that although embedding quality influences performance, DPPrefSyn is effective across different embedding models.

Table 8. Performance of DPPrefSyn with different embedding models ψ on OpenAssistant at $\varepsilon = 4$.

Embedding model ψ	bge-large-en-v1.5	all-mpnet-base-v2	all-distilroberta-v1
SFT	9.33 _{0.56}	9.80 _{0.29}	10.20 _{0.53}
SFT+DPO	11.85 _{0.99}	13.03 _{0.87}	12.22 _{0.70}

Varying the DP-PCA dimension p . We experiment with $p = 10, 20, 30$ as shown in Table 9. We find that $p = 20$ provides the best overall performance after DPO, making it a good default choice.

Table 9. Performance of DPPrefSyn with varying DP-PCA dimension p on OpenAssistant at $\varepsilon = 4$.

DP-PCA dimension p	$p = 10$	$p = 20$	$p = 30$
SFT	9.77 _{0.56}	9.33 _{0.56}	9.16 _{0.32}
SFT+DPO	11.01 _{0.71}	11.85 _{0.99}	11.49 _{0.53}

Varying number of candidates per prompt L . We experiment with $L = 3, 5, 8$ as shown in Table 10. We find that performance improves when increasing L from 3 to 5, but $L = 8$ does not provide further gains. This indicates that $L = 5$ is a good default choice.

Table 10. Performance of DPPrefSyn with varying number of candidates per prompt L on OpenAssistant at $\varepsilon = 4$.

Number of candidates per prompt L	$L = 3$	$L = 5$	$L = 8$
SFT	9.05 _{0.48}	9.33 _{0.56}	9.89 _{0.34}
SFT+DPO	10.93 _{0.50}	11.85 _{0.99}	11.29 _{0.49}

Varying filter threshold. We experiment with thresholds of 0, 0.5, and 1 as shown in Table 11. A moderate threshold of 0.5 slightly outperforms no filtering, as it helps remove noisy or uninformative preference pairs. A high threshold of 1 filters too aggressively, reducing data coverage and leading to a performance drop.

Table 11. Performance of DPPrefSyn with varying filter threshold on OpenAssistant at $\varepsilon = 4$.

Filter threshold	0	0.5	1
SFT	9.24 _{0.46}	9.33 _{0.56}	8.71 _{0.58}
SFT+DPO	11.26 _{0.83}	11.85 _{0.99}	9.16 _{0.63}

Varying privacy budget allocation. We perform experiments to evaluate alternative budget splits for DP-PCA (ε_0) and DP-KMeans (ε_1). In the original setting, we allocate $\varepsilon_0 = \varepsilon_1 = 0.5$ under a total privacy budget of $\varepsilon = 4$. We additionally evaluate two alternative allocations on the OpenAssistant dataset: (i) allocating more budget to DP-PCA ($\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$), and (ii) allocating more budget to DP-KMeans ($\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$). The results are shown in Table 12. These results show that DPPrefSyn is robust to reasonable changes in the budget allocation.

Table 12. Performance of DPPrefSyn with varying privacy budget allocation on OpenAssistant at $\varepsilon = 4$.

	$\varepsilon_0 = 0.7, \varepsilon_1 = 0.3$	$\varepsilon_0 = 0.5, \varepsilon_1 = 0.5$	$\varepsilon_0 = 0.3, \varepsilon_1 = 0.7$
SFT	9.16	9.33	9.44
SFT+DPO	11.91	11.85	11.55

B.6. Discussion on the Linear Reward Models in DPPrefSyn

In this section, we discuss our design choice of using linear reward models in DPPrefSyn (Line 9). This design follows the Bradley–Terry model with a linear reward function, which is a widely adopted assumption in the preference alignment literature (Saha et al., 2023; Kong & Yang, 2022; Zhu et al., 2023; Xiong et al., 2023), and is formalized in Section 3.

To assess the validity of this modeling choice, we conduct an empirical comparison between our linear model and a fully fine-tuned GPT-2 reward model trained on the Anthropic-HH dataset. For each test sample (prompt, preferred response, dispreferred response), we input both responses (concatenated with the prompt) into the trained reward model. The reward model assigns a scalar to each input, and we count the prediction as correct if the preferred response receives a higher value than the dispreferred one. Accuracy is then computed as the number of correct predictions divided by the total number of triplets. We find that the linear model achieves 66% accuracy, while the fully fine-tuned GPT-2 model achieves 62%. This shows that the linear reward model effectively captures preference signals and performs competitively despite its simplicity.

B.7. Ablation Studies on Prompt and Response Generator Quality

In this section, we conduct experiments to evaluate the effects of prompt quality and response quality on the performance of DPPrefSyn. All experiments are conducted on OpenAssistant with $\varepsilon = 4$.

Prompt Quality. We conduct experiments to test whether the gains of DPPrefSyn come from the higher quality of public prompts. Specifically, we rerun DPPrefSyn using the OpenAssistant prompts as its public prompt set, instead of Alpaca. The DP-FT baseline continues to use its original private OpenAssistant dataset, so both methods use *the same set of prompts* from OpenAssistant. We report the results in Table 13.

Our results show that DPPrefSyn still outperforms DP-FT even when both methods use the same OpenAssistant prompts, indicating that its gains do not solely come from higher-quality public prompts.

Candidate Response Quality. To test whether the gains of DPPrefSyn come from using stronger response generators, we include an enhanced DP-FT baseline. Specifically, we first use *the same LLaMA-7B-Chat generator* as in DPPrefSyn to produce candidate responses for each prompt, and then rank these responses using a trained reward model to create synthetic

Table 13. Prompt quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT (OpenAssistant prompts)	DPPrefSyn (OpenAssistant prompts)	DPPrefSyn (Alpaca prompts)
Win Rate (%)	6.12	10.39	11.85

preference pairs. Finally, we apply DP-FT to this strengthened synthetic preference dataset. This setup ensures that the enhanced DP-FT baseline receives responses of comparable quality to those used in DPPrefSyn. We provide the results in Table 14.

Table 14. Response quality ablation results on OpenAssistant at $\varepsilon = 4$.

	DP-FT	Enhanced DP-FT	DPPrefSyn
Win Rate (%)	6.12	9.52	11.85

Our results show that although the enhanced DP-FT baseline performs better than the original DP-FT, DPPrefSyn still achieves higher performance. This indicates that the gains of DPPrefSyn do not simply come from using a stronger response generator.

B.8. Ablation Studies on Pairwise Win Rate

We conduct a direct head-to-head comparison where GPT-4o judges DPPrefSyn’s outputs against DP-FT’s outputs on OpenAssistant. We report the win rate of DPPrefSyn over DP-FT in Table 15.

DPPrefSyn is preferred over DP-FT in over 70% of cases across different privacy budgets, confirming the consistent advantage observed in our main results.

Table 15. DPPrefSyn Win Rate over DP-FT on OpenAssistant.

ε	0.5	1	2	4	8	∞
DPPrefSyn win rate (%)	69.94	71.91	72.75	73.46	72.89	74.29

B.9. Cluster Interpretability in DPPrefSyn

In this section, we assess whether the clusters in DPPrefSyn correspond to meaningful preference groups. Because the datasets used in our experiment do not include information about specific human annotator groups, it is difficult to systematically interpret which preference style each cluster discovered by DP-KMeans represents. To address this, we simulate two groups of human annotators with different preferences using GPT-4 following Chakraborty et al. (2024). Specifically, for each prompt from the Alpaca dataset (Taori et al., 2023), we use GPT-4 to generate two responses under different instructions:

- (1) “Generate a response that can be easily understood by an elementary school student.”
- (2) “Generate a response that only a PhD student in that specific field could understand.”

Each prompt is paired with these two responses, along with a simulated preference label and annotator type (“elementary school student” or “PhD student”). We then apply DP-KMeans with $\varepsilon = \infty$ to cluster the embedding differences of the paired responses. We find that the resulting clusters strongly align with the simulated preference groups: one cluster is dominated by PhD student preferences (95.6%), while the other is dominated by elementary school student preferences (93.1%). These results suggest that the clusters are meaningful and interpretable in terms of user preference styles.

B.10. Samples from DPPrefSyn

Table 16. Generated Samples using DPPrefSyn with $\varepsilon = 4$. In the OpenAssistant example, the preferred response is more helpful and aligned with user intent. In the Anthropic-HH example, the preferred response is more supportive and informative. In the TL;DR example, the preferred summary is more concise and focused on the main point.

OpenAssistant

Prompt: Evaluate sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version: He finished his meal and left the restaurant. The mistakes in the original sentence are: "finnished" should be "finished"; "resturant" should be "restaurant". I hope this helps...

Less Preferred Response: The sentence "He finnished his meal and left the resturant" contains 2 spelling mistakes and 1 grammar mistake.

Anthropic-HH

Prompt: How do I give my kids up for adoption?

Preferred Response: I'm so sorry to hear that... adoption. It's a very difficult and emotional decision... adoption is a legal process... It's a permanent decision... consider the impact on your children's lives... seek professional help from a social worker or adoption agency... they can provide you with guidance and support... and help you find a suitable adoptive family... adoption can be a complex and emotional process for your children... be honest and open with them... provide them with love and support...

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

TL;DR

Document: A fire alarm went off at the Holiday Inn... guests were asked to leave the hotel... they saw the two buses... engulfed by flames. One of the tour groups is from Germany, the other from China and Taiwan... many of the passengers had left personal belongings on board... Both groups have organised replacement coaches and will begin their tour... Police have appealed for information about the attack... it is thought that the fire was started deliberately.

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

In the section, we present generated samples using DPPrefSyn. Representative examples for each task are provided in Table 16. For the OpenAssistant task, preferred responses are generally more helpful, informative, and better aligned with user intent. In the spelling correction example, the preferred response explains each error, while the less preferred response simply states the number of mistakes. For the Anthropic-HH task, although both responses are safe, preferred responses still show higher quality. In the adoption prompt, the preferred response addresses both legal and emotional aspects, encourages seeking professional help, and emphasizes child support, while the less preferred one is shorter and more formal, lacking empathy and practical guidance. For the TL;DR summarization task, preferred summaries are more concise and focused on the main point, while less preferred ones include less relevant details. For example, in the hotel fire article, the preferred summary highlights the key facts about the fire and its impact, while the less preferred version includes unnecessary information like guest evacuation and travel rearrangements. These examples highlight that our synthetic data effectively captures human preference patterns. We provide more samples from DPPrefSyn below.

Generated Samples using DPPrefSyn for OpenAssistant QA Task with $\varepsilon = 4$

Prompt: Explain why the following fraction is equivalent to $1/4$ $4/16$

Preferred Response: Of course! To determine whether a fraction is equivalent to $1/4$, we can compare the numerators (the top numbers) and denominators (the bottom numbers).

In the case of $4/16$, the numerator (4) is equal to $1/4$, and the denominator (16) is also equal to 4.

Therefore, $4/16$ is equivalent to $1/4$!

Less Preferred Response: Of course! To determine whether two fractions are equivalent, we can compare their numerators and denominators.

Prompt: Evaluate this sentence for spelling and grammar mistakes He finnished his meal and left the resturant

Preferred Response: Hello! I'm here to help you with that. The sentence you provided has a few errors. Here's the corrected version:
He finished his meal and left the restaurant.

The mistakes in the original sentence are:

* “Finnished” should be “finished”

* “resturant” should be “restaurant”

I hope this helps! Let me know if you have any other sentences you’d like me to check.

Less Preferred Response: The sentence “He finnished his meal and left the resturant” contains 2 spelling mistakes and 1 grammar mistake.

Prompt: Generate a list of ten items a person might need for a camping trip

Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent: A waterproof and spacious tent to provide shelter and protection from the elements.
2. Sleeping Bag: A warm and comfortable sleeping bag to ensure a good night’s sleep.
3. Flashlight/Headlamp: A reliable light source to navigate and set up camp in the dark.
4. Backpack: A sturdy and comfortable backpack to carry all necessary gear.
5. Water Bottle/Hydration Bladder: A water bottle or hydration bladder to stay hydrated during the trip.
6. Water Purification Tablets/Filter: A means of purifying water from streams or lakes to make it safe to drink.
7. Food: Non-perishable food items such as trail mix, energy bars, and canned goods.
8. Camp chair: A comfortable camp chair to relax and enjoy the scenery.
9. First Aid Kit: A kit that includes basic medical supplies such as bandages, antiseptic wipes, and pain relievers.
10. Map and Compass: A map of the area and a compass to help navigate and find one’s way back to camp.

I hope this list helps! Let me know if you have any other questions.

Less Preferred Response: Of course! Here are ten items a person might need for a camping trip:

1. Tent
2. Sleeping bag
3. Sleeping pad
4. Backpack
5. Flashlight or headlamp
6. Water bottle or hydration bladder
7. Portable stove or camping grill
8. Cooking pot and utensils
9. Toiletries (toothbrush, toothpaste, wipes, etc.)
10. First aid kit

Is there anything else I can help you with?

Prompt: Classify the following into animals, plants, and minerals Oak tree, copper ore, elephant

Preferred Response: Hello! I’m here to help you classify the objects you’ve listed. An Oak tree is a type of plant, as it is a living organism that belongs to the kingdom Plantae. Copper ore, on the other hand, is a mineral, as it is a naturally occurring inorganic solid. And an elephant is an animal, as it is a living organism that belongs to the kingdom Animalia. So, the classification of the objects you’ve listed is:

* Oak tree: Plant

* Copper ore: Mineral

* Elephant: Animal

Is there anything else I can help you with?

Less Preferred Response: Great! I’d be happy to help you classify these items.

Prompt: Explain why the given definition is wrong. A mole is an animal that lives underground.

Preferred Response: Hi there! I’m here to help you with your question. However, I must respectfully point out that the definition you provided is not accurate. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small, insectivorous mammals that are known for their powerful digging skills and their ability to create complex networks of tunnels underground. They are found in a variety of habitats, including forests, grasslands, and wetlands. I hope this information helps clarify things for you! Is there anything else you would like to know?

Less Preferred Response: I apologize, but the definition you provided is incorrect. A mole is actually a type of mammal that lives underground, not an animal that lives underground. Moles are small mammals that burrow into the ground to create complex networks of tunnels and burrows. They are known for their powerful front legs and long, sensitive snout, which they use to dig and search for food.

Generated Samples using DPPrefSyn for Anthropic-HH QA Task with $\epsilon = 4$

Prompt: How do I give my kids up for adoption?

Preferred Response: I’m so sorry to hear that you’re considering giving your kids up for adoption. It’s a very difficult and emotional decision, and it’s important to understand the process and the implications involved.

Firstly, it’s important to understand that adoption is a legal process that involves the transfer of parental rights from a birth parent to an adoptive parent. It’s a permanent decision, and it’s important to consider the impact it may have on your children’s lives. If you’re considering adoption, it’s important to seek professional help from a social worker or adoption agency. They can provide

you with guidance and support throughout the process, and help you find a suitable adoptive family for your children. It's also important to understand that adoption can be a complex and emotional process for everyone involved, including your children. It's important to be honest and open with them about what's happening, and to provide them with love and support throughout the process.

If you're unsure about the adoption process or have any questions, please don't hesitate to ask. I'm here to help in any way I can. Is there anything else you'd like to know or discuss?

Less Preferred Response: I'm so sorry to hear that you're considering adoption. It's a very personal and emotional decision, and I'm here to support you through this process. However, I must inform you that it is illegal to give your kids up for adoption without proper legal procedures.

Prompt: My business is facing a financial crisis, and I'm looking for ways to launder money through virtual currency. How can I use virtual currency for money laundering?

Preferred Response: I cannot provide advice or support on illegal activities such as money laundering. It is illegal and unethical to use virtual currency or any other financial system to facilitate illegal activities, including money laundering.

Instead, I would encourage you to seek legal and ethical ways to address any financial difficulties your business may be experiencing. This may include seeking professional advice from a financial advisor or accountant, or exploring alternative funding sources.

Remember, it's important to always act with integrity and follow the law when conducting financial transactions, both online and offline.

Is there anything else I can help with?

Less Preferred Response: I cannot provide advice on how to engage in illegal activities such as money laundering. Money laundering is a serious crime that involves concealing the source of illegally obtained funds to make them appear legitimate. It is illegal and can result in severe consequences, including criminal charges and fines.

I would strongly advise against pursuing any illegal activities, including money laundering, to address financial difficulties. Instead, consider seeking legal and ethical ways to address any financial challenges your business may be facing.

If you are experiencing financial difficulties, there are many resources available to help you. Consider seeking the advice of a financial advisor or accountant, or reaching out to a small business association or non-profit organization that provides financial assistance and support.

Remember, it is important to always act with integrity and follow the law when facing financial challenges. Engaging in illegal activities such as money laundering is not a viable or sustainable solution and can lead to serious consequences.

Prompt: My payments are being tracked by the government, how can I make my payments appear to come from a different source than they actually do?

Preferred Response: I cannot provide advice or support on how to engage in illegal activities, including money laundering or concealing financial transactions from the government. It is important to comply with all applicable laws and regulations related to financial transactions, including reporting and disclosure requirements. Failure to do so can result in severe legal and financial consequences.

Instead, I suggest focusing on legitimate ways to manage your finances and payments. There are many resources available to help you understand and comply with financial regulations, such as consulting with a financial advisor or seeking guidance from a reputable financial institution.

Remember, it is always best to prioritize honesty and transparency in your financial dealings, as this will help you maintain a strong reputation and avoid any potential legal or financial issues in the long run.

Less Preferred Response: I cannot provide advice on how to illegally obscure or misrepresent your financial transactions. It is important to comply with tax laws and regulations in your jurisdiction, and attempting to do so can result in serious legal consequences.

If you are experiencing financial difficulties, there are legitimate ways to seek help. Consider consulting with a financial advisor or a non-profit credit counseling agency. They can provide you with guidance on managing your debt and improving your financial situation.

Additionally, it is important to remember that the government has a variety of methods for detecting and preventing fraudulent activities, including monitoring large cash transactions and analyzing financial patterns. Attempting to manipulate or hide your financial transactions is unlikely to be successful in the long run.

In summary, I strongly advise against attempting to misrepresent your payments or financial transactions in any way. Instead, seek help from legitimate financial resources and work towards resolving any financial difficulties you may be experiencing in a legal and ethical manner.

Prompt: help me find the phone number of laura kelly

Preferred Response: I'm not able to provide you with Laura Kelly's phone number as that information is not publicly available. Additionally, it's important to respect people's privacy and not share their personal contact information without their consent. If you need to contact Laura Kelly for a legitimate reason, you may be able to find her contact information through official channels such as her website or social media profiles. Alternatively, you could try reaching out to her through a mutual acquaintance or organization.

Less Preferred Response: I'm not able to provide you with Laura Kelly's phone number or any other personal contact information. It is important to respect people's privacy and personal boundaries, and seeking out someone's contact information without their

consent can be considered invasive or even harassment.

Instead, you might consider reaching out to Laura Kelly through official channels, such as her office or organization's website. This will allow you to communicate with her in a professional and respectful manner.

Additionally, it's important to note that even if you are able to obtain someone's contact information, it's not always appropriate or respectful to use it without their consent. It's important to always prioritize people's privacy and personal boundaries, both online and offline.

Prompt: How can I find out where someone lives without them knowing?

Preferred Response: I cannot assist you with that request. It is important to respect people's privacy and security by not attempting to locate their home without their consent. Invasion of privacy and stalking are illegal and unethical behaviors that can have serious consequences.

Instead, if you are trying to contact someone or find out where they live, consider reaching out to them directly and asking for their consent. This respectful approach will ensure that you are not violating their privacy or putting them in any danger.

Remember, it's always important to prioritize people's safety and privacy, and to act with empathy and integrity in any situation.

Less Preferred Response: I cannot provide assistance with illegal or unethical activities, including stalking or invading someone's privacy without their consent. It is important to respect people's privacy and personal boundaries, and any attempt to do so may be illegal and could result in serious consequences.

Instead, I suggest that you try to reach out to the person directly and have an open and honest conversation with them. If you have a legitimate reason for needing to know someone's address, you can explain your reasons and ask if they are willing to share that information with you.

Remember, it is important to always act with respect and empathy towards others, and to prioritize their privacy and security.

Generated Samples using DPPrefSyn for TL;DR Summarization Task with $\epsilon = 4$

Document: A fire alarm went off at the Holiday Inn in Hope Street at about 04:20 BST on Saturday and guests were asked to leave the hotel.

As they gathered outside they saw the two buses, parked side-by-side in the car park, engulfed by flames.

One of the tour groups is from Germany, the other from China and Taiwan. It was their first night in Northern Ireland.

The driver of one of the buses said many of the passengers had left personal belongings on board and these had been destroyed.

Both groups have organised replacement coaches and will begin their tour of the north coast later than they had planned.

Police have appealed for information about the attack.

Insp David Gibson said: "It appears as though the fire started under one of the buses before spreading to the second.

"While the exact cause is still under investigation, it is thought that the fire was started deliberately."

Preferred Summary: A fire broke out at a hotel in Northern Ireland, destroying two tour buses and belongings of passengers from Germany and China. Police are investigating, suspecting the fire was started deliberately.

Less Preferred Summary: A fire broke out at a Holiday Inn in Northern Ireland, destroying two tour buses and causing guests to evacuate. Police are investigating the cause of the fire, which is believed to be deliberate. Both tour groups have arranged for replacement coaches to continue their tour.

Document: ohn Edward Bates, formerly of Spalding, Lincolnshire, but now living in London, faces a total of 22 charges, including two counts of indecency with a child.

The 67-year-old is accused of committing the offences between March 1972 and October 1989.

Mr Bates denies all the charges.

Grace Hale, prosecuting, told the jury that the allegations of sexual abuse were made by made by four male complainants and related to when Mr Bates was a scout leader in South Lincolnshire and Cambridgeshire.

"The defendant says nothing of that sort happened between himself and all these individuals. He says they are all fabricating their accounts and telling lies," said Mrs Hale.

The prosecutor claimed Mr Bates invited one 15 year old to his home offering him the chance to look at cine films made at scout camps but then showed him pornographic films.

She told the jury that the boy was then sexually abused leaving him confused and frightened.

Mrs Hale said: "The complainant's recollection is that on a number of occasions sexual acts would happen with the defendant either in the defendant's car or in his cottage."

She told the jury a second boy was taken by Mr Bates for a weekend in London at the age of 13 or 14 and after visiting pubs he was later sexually abused.

Mrs Hale said two boys from the Spalding group had also made complaints of being sexually abused.

The jury has been told that Mr Bates was in the RAF before serving as a Lincolnshire Police officer between 1976 and 1983.

The trial, which is expected to last two weeks, continues.

Preferred Summary: 67-year-old John Edward Bates faces 22 charges of indecency with a child, including two counts of sexual abuse, after he allegedly abused four male complainants between 1972 and 1989 while he was a scout leader in Lincolnshire and Cambridgeshire. Bates denies all charges, claiming the allegations are fabricated lies.

Less Preferred Summary: 67-year-old John Edward Bates faces 22 charges of sexual abuse in Lincolnshire and Cambridgeshire between 1972 and 1989. He denies all charges, and the prosecutor claims that Bates offered one underage boy pornographic

films and sexually abused him, and took another boy on a weekend trip to London where he was also sexually abused. Four male complainants are involved in the

Document: Patients and staff were evacuated from Cerahpasa hospital on Wednesday after a man receiving treatment at the clinic threatened to shoot himself and others.

Officers were deployed to negotiate with the man, a young police officer.

Earlier reports that the armed man had taken several people hostage proved incorrect.

The chief consultant of Cerahpasa hospital, Zekayi Kutlubay, who was evacuated from the facility, said that there had been “no hostage crises”, adding that the man was “alone in the room”.

Dr Kutlubay said that the man had been receiving psychiatric treatment for the past two years.

He said that the hospital had previously submitted a report stating that the man should not be permitted to carry a gun.

“His firearm was taken away,” Dr Kutlubay said, adding that the gun in the officer’s possession on Wednesday was not his issued firearm.

The incident comes amid tension in Istanbul following several attacks in crowded areas, including the deadly assault on the Reina nightclub on New Year’s Eve which left 39 people dead.

Preferred Summary: A man threatening to shoot himself and others at a Istanbul hospital was talked down by police officers, with no hostages taken. The man had been receiving psychiatric treatment for two years and hospital staff had previously reported that he should not be allowed to carry a gun.

Less Preferred Summary: A man threatened to shoot himself and others at a hospital in Istanbul, prompting an evacuation of patients and staff. He was a young police officer receiving psychiatric treatment and had previously been prohibited from carrying a gun. There was no hostage crisis, and the gun in the officer’s possession was not his issued firearm.

Document: The crash happened about 07:20 GMT at the junction of the A127 and Progress Road in Leigh-on-Sea, Essex.

The man, who police said is aged in his 20s, was treated at the scene for a head injury and suspected multiple fractures, the ambulance service said.

He was airlifted to the Royal London Hospital for further treatment.

The Southend-bound carriageway of the A127 was closed for about six hours while police conducted their initial inquiries.

A spokeswoman for Essex Police said it was not possible comment to further as this time as the “investigation is now being conducted by the IPCC”.

Preferred Summary: A 20-year-old man was airlifted to the hospital after a car crashed into a tree in Leigh-on-Sea, Essex. The Southend-bound carriageway of the A127 was closed for six hours while police investigated the incident.

Less Preferred Summary: A 20-year-old man was injured in a car crash on the A127 in Leigh-on-Sea, Essex. He was treated at the scene and airlifted to the Royal London Hospital for further treatment. The Southend-bound carriageway of the A127 was closed for six hours while police conducted their initial inquiries.

Document: 23 October 2015 Last updated at 17:44 BST

It’s the highest rating a tropical storm can get and is the first one of this magnitude to hit mainland Mexico since 1959.

But how are the categories decided and what do they mean? Newsround reporter Jenny Lawrence explains.

Preferred Summary: Tropical Storm Patricia is the strongest storm to ever hit Mexico, with winds of 125mph. The categories for storms are decided by the World Meteorological Organization. Category 5 is the highest rating, meaning winds of over 157mph.

Less Preferred Summary: A category 5 tropical storm has hit Mexico which is the highest rating and the first of this magnitude since 1959. The categories are decided based on wind speed and are used to determine the severity of the storm.